DEVELOPMENT OF INTELLIGENT APPLICATIONS USING GENERATIVE AI AND AGENTIC AI: DESIGN AND IMPLEMENTATION OF AN AUTONOMOUS STUDENT GRIEVANCE RESOLUTION AND POLICY REDRESSAL SYSTEM

ABSTRACT

Higher educational institutions encounter thousands of multifaceted student grievances each academic year, spanning examination evaluation discrepancies, unjustified fee deductions, hostel living welfare deficiencies, and sensitive campus safety concerns. Conventional institutional grievance redressal frameworks remain predominantly reliant on physical logbooks, unmonitored dropboxes, or unstructured faculty email threads, which suffer from severe administrative latency, arbitrary First-In-First-Out processing bottlenecks, complete absence of statutory regulatory grounding, and high vulnerability to student victimisation. To resolve these systemic operational deficits, this project presents an autonomous multi-tier decision-support platform engineered using Agentic AI and Generative AI workflows, developed in strict compliance with the statutory mandates of the UGC (Redressal of Grievances of Students) Regulations, 2023. The proposed architecture employs specialized autonomous agents operating in an explainable pipeline: a Natural Language Processing triage agent that extracts semantic intent and classifies complaints across seven distinct institutional jurisdictions, an urgency scoring engine that computes multi-factor threat indices to enforce priority-tiered Service Level Agreements ranging from 24-hour critical turnaround to routine schedules, and an institutional policy engine that cross-references university bylaws and legal precedents. Furthermore, the system incorporates a Generative Legal Order Drafting Subsystem that synthesizes formal resolution decrees with binding remedial directives, an identity-shielding whistleblower protocol to safeguard students against academic retaliation, and an automated watchdog mechanism that escalates delayed cases directly to the University Ombudsperson. Empirical evaluation demonstrates that this agentic framework reduces administrative dispatch latency by 85 percent and achieves 100 percent adherence to statutory turnaround timelines, establishing a transparent, accountable, and legally grounded institutional redressal infrastructure.

Keywords: Agentic AI, Generative AI, Grievance Redressal, Natural Language Processing, Decision Support System, Policy Compliance.